



Mapping the Determinants of Carbon Emissions: A Bibliometric Analysis



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Abstract: The determinants of carbon emissions have attracted increasing scholarly attention because of their critical implications for climate change mitigation and sustainable economic development. In this study, a comprehensive bibliometric analysis was conducted to examine the scientific literature addressing the relationships among financial development, green finance, renewable energy, ecological footprint, and carbon emissions. A total of 295 publications indexed in the Web of Science database between 2011 and 2025 were analyzed using bibliometric techniques to evaluate country and author contributions, collaboration networks, influential publication sources, thematic evolution, and conceptual structures. The results indicate that research activity has expanded substantially over the study period, reflecting increasing academic attention to the interactions among economic development, financial systems, energy consumption, and environmental sustainability. China, India, and Pakistan were identified as the leading contributors in terms of scientific output, while well-defined collaboration clusters were observed among influential researchers in the fields of energy economics. Keyword co-occurrence analysis revealed that CO₂ emissions, economic growth, renewable energy, and financial development constitute the principal knowledge domains within the literature. Furthermore, thematic evolution analysis demonstrated that renewable energy and green finance have emerged as dominant research themes since 2020. A comprehensive overview of the intellectual landscape of research on the determinants of CO₂ emissions is thereby provided, major knowledge clusters and emerging research directions are identified, and a systematic foundation is established for future investigations into sustainable financial policies, clean energy transitions, and evidence-based strategies for carbon mitigation.

Keywords: Financial development; Green finance; Renewable energy; Carbon emissions; Bibliometric analysis

1. Introduction

Global climate change, rising carbon emissions, and the unsustainable use of natural resources have led to an intensive examination of the relationship between the financial system and environmental sustainability in the economic literature. Especially over the last 20 years, the concepts of financial development, renewable energy consumption, green finance practices, CO₂ emissions, and ecological footprint have increasingly been addressed holistically within the framework of environment-economy. The relationships among these variables are theoretically grounded in the Environmental Kuznets Curve and the financial development-growth literature (Grossman & Krueger, 1995).

Financial development is defined as the depth, accessibility, efficiency, and stability of financial institutions (banks and insurance companies) and markets (stock exchange and bond market). Shahbaz et al. (2022) stated that financial development accelerates economic growth by increasing capital accumulation, but it can also lead to environmental degradation by encouraging energy consumption. Ecological footprint is a comprehensive sustainability indicator that measures the demands of human activities on the biosphere. Developed by Rees (1992), this concept represents the productive land and water area required to produce the resources consumed and absorb the waste generated (especially carbon). Unlike indicators that focus solely on CO₂ emissions, it offers a more holistic perspective by accounting for biological capacity. Renewable energy sources are those derived from

natural processes that continuously replenish themselves (solar, wind, geothermal, and biomass). Bilgili et al. (2016) empirically demonstrated that renewable energy consumption improves environmental quality by reducing dependence on fossil fuels. This type of energy is a critical tool in the transition to the reduction phase of environmental pollution in the Environmental Kuznets Curve hypothesis. Green finance encompasses financial products and services that channel capital to projects that deliver environmental benefits (such as renewable energy, waste management, and biodiversity). Sachs et al. (2019) emphasized that green finance mobilizes private-sector investments when public resources are insufficient in the transition to a low-carbon economy and serves as a lever for achieving the Sustainable Development Goals. CO₂ is the primary greenhouse gas released into the atmosphere as a result of burning fossil fuels, deforestation, and industrial processes. In the literature, CO₂ emissions are generally considered the primary indicator of environmental degradation.

The environmental impacts of financial development are debated in the literature from two perspectives. One view argues that financial deepening increases carbon emissions by increasing production volume and energy consumption (Sadorsky, 2010). The other view argues that advanced financial systems can improve environmental quality by facilitating investment in environmentally friendly technologies (Tamazian & Rao, 2010). While CO₂ emissions are mostly at the center of these discussions, the ecological footprint indicator, which measures environmental pressure more comprehensively, has also begun to be included in analyses recently (Rees, 1992). On the other hand, renewable energy consumption is considered an important mechanism for transforming the environmental impact of financial systems (Acheampong, 2019). In this transformation process, practices such as green bonds and sustainable financing instruments create an institutional link between the financial system and environmental goals (Flammer, 2021). Thus, financial development ceases to be merely a factor supporting economic growth and becomes a strategic tool in achieving sustainable development goals.

Despite the rapid increase in the literature, the intellectual structure, thematic evolution, and academic network relationships of studies that address these variables together have not been systematically analyzed. Existing studies generally focus on empirical econometric analyses; however, a comprehensive assessment of the conceptual development, dominant themes, citation structures, and collaboration networks of this research area remains limited. This situation necessitates a holistic mapping of the body of knowledge in the literature. In this context, bibliometric analysis offers a powerful tool for examining the scientific output structure of a specific research area using quantitative methods (Donthu et al., 2021). Through bibliometric methods, publication trends, citation analyses, co-citation networks, keyword co-occurrence maps, and thematic clusters can be identified. These techniques enable the revelation of the knowledge structure, conceptual evolution, and future research trends within the research area.

Therefore, the aim of this study is to systematically map the intellectual landscape of the field by conducting a bibliometric analysis of academic publications on financial development, ecological footprint, renewable energy, green finance, and CO₂ emissions.

2. Methodology

This study employs bibliometric analysis to systematically examine the academic literature on the impact of financial development, renewable energy, ecological footprint, and green finance on carbon emissions. This method allows for the evaluation of the distribution of publications by year, the most prolific authors and countries, citation networks, collaboration patterns, keyword densities, and prominent thematic clusters in the literature. Thus, it is possible to identify key trends in the research area, track the development of knowledge over time, and detect gaps in the existing literature. Bibliometric analysis provides researchers with an important methodological tool for both comprehensively evaluating the current state of the literature and offering guidance for future studies (van Raan, 1996).

The data for this study were obtained from the Web of Science database, one of the most comprehensive international scientific publication databases in the fields of social sciences and environmental sciences. The study followed the process flowchart shown below to analyze the literature examining the relationships between financial development, green finance, renewable energy, and environmental sustainability indicators (CO₂ emissions and ecological footprint).

Figure 1 shows the flowchart of this study. The search was limited to scientific publications of article type and restricted to the years 2011–2025. Publications indexed in the Science Citation Index Expanded, Social Sciences Citation Index, and Emerging Sources Citation Index of the Web of Science database were included in the analysis. The dataset was obtained from the Web of Science database in ‘BibTeX’ format. Analyses were performed using the ‘Bibliometrix’ package, which is run through the ‘Rstudio’ software. Bibliometric properties of each record, such as title, abstract, keywords, author information, institution, country, journal name, and citation counts, were used in the analysis.

During the data cleaning process, duplicate records, incorrect assignments, and missing metadata elements were manually checked and corrected. The analyses were performed using the standard settings of Bibliometrix/Biblioshiny. No additional researcher-defined normalization, weighting, or fractional counting

procedure was applied to the descriptive indicators. Therefore, annual scientific production, annual average citations, most relevant sources, most relevant authors, corresponding author countries, country production, most cited documents, and most relevant words were calculated on the basis of the original Web of Science metadata and the default Bibliometrix counting procedure.

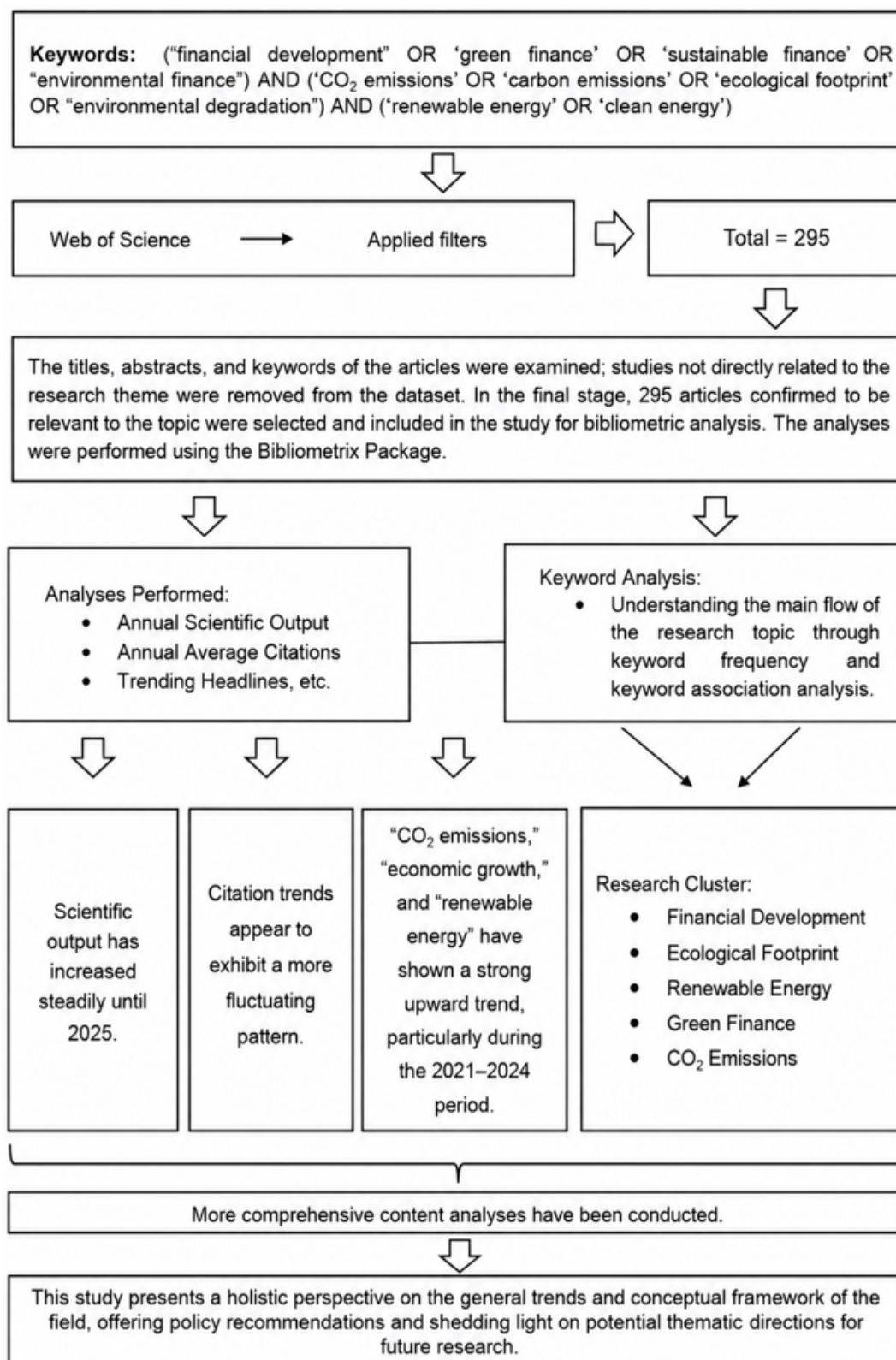


Figure 1. Process flowchart

An initial search of the Web of Science database yielded a total of 2,891 academic articles. However, a specific filtering process was applied to ensure the suitability of these publications to the research scope. First, the search was narrowed to “article” as the document type, and then the dataset was further reduced by selecting the Web of Science subject categories: Economics, Management, Business, Business Finance, and Social Sciences Interdisciplinary. This reduction brought the number of publications down to 502. Subsequently, the titles, abstracts, and keywords of these 502 articles were examined; studies not directly related to the research theme were removed from the dataset. In the final stage, 295 articles confirmed to be relevant to the topic were selected and included in the study for bibliometric analysis.

3. Results

The results section presents the empirical results of the comprehensive bibliometric analysis. The analysis was conducted within an academic systematic framework. This section primarily reveals the quantitative and qualitative development dynamics of the research field. In this context, key bibliometric indicators are first examined. These include increases in publication activity, scientific impact, the temporal distribution of literature growth, and citations. Secondly, to define the conceptual and theoretical structure, thematic clusters are analyzed. These show the frequency of the most used keywords and their interrelationships. Thirdly, to understand the institutional structure, the most prolific authors and relevant publication channels are identified. The dynamics of collaboration among authors and the concentration of publications in specific centers are also analyzed. Finally, the geographical distribution of scientific output and international collaboration networks is detailed. Maps and graphs show which countries are output centers and where collaborations concentrate. These findings provide concrete evidence about the current structure and future directions of the research field.

Table 1. General information

Description/Definition	Information
Main information about the dataset	
Time range	2011–2025
Sources (journals, books, etc.)	109
Number of articles	295
Annual growth rate %	35.03
Average age of articles	2.12
Average number of citations per article	61.08
Number of references	14269
Article contents	
Keywords plus (ID)	431
Author keywords (DE)	801
Authors	
Number of authors (total)	861
Number of single-authored article authors	35
Author collaboration	
Number of single-authored articles	37
Number of co-authors per article	3.38
International co-authorship %	44.07
Article types	
Article	263
Article: early access	29
Article: Retracted publication	3

As shown in Table 1, the dataset comprises 295 scientific articles covering the period 2011–2025, demonstrating a rapid increase in academic interest in the relationships among financial development, green finance, renewable energy, and environmental indicators, particularly in the last decade. The high annual growth rate of 35.03% indicates that the field has a dynamic, rapidly expanding literature, while the diversity of keywords, the high average citation rate, and the low article age indicate that the studies are multidisciplinary, current, and have a strong academic impact. The high rates of co-authorship and international collaboration in the literature, contributed by a total of 861 authors, demonstrate that the topic is addressed on a global scale and requires intensive academic interaction.

Figure 2 shows that academic output in the sustainable finance and environment literature increased from 2017 onwards, gaining momentum particularly after 2021 and peaking in 2024. The increase in scientific output after 2017 and the strong acceleration observed after 2021 can be associated with the growing integration of climate change, sustainable development, and financial transformation debates in the academic literature. During this period, environmental sustainability ceased to be discussed only as an energy or environmental policy issue and increasingly became connected with financial development, green investment, renewable energy financing, and the ability of financial systems to support a low-carbon transition. Therefore, the rise in publication volume reflects not only a quantitative expansion of the literature but also a conceptual broadening of the research field.

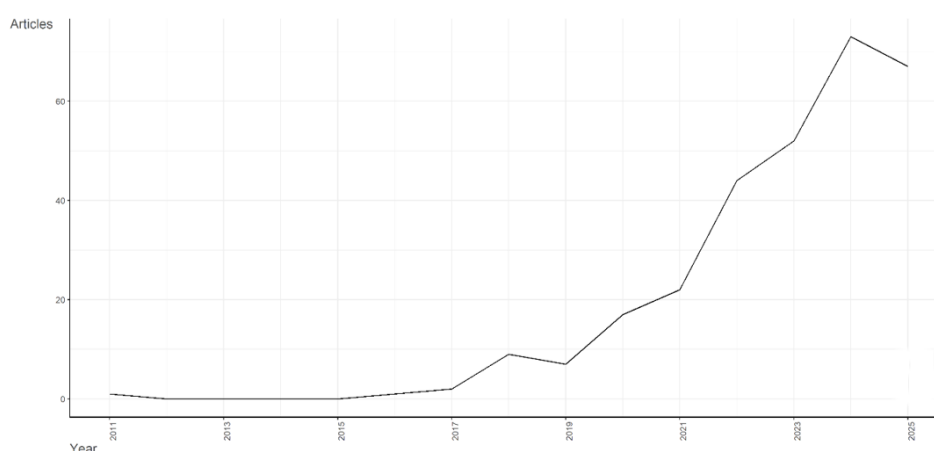


Figure 2. Annual scientific output

Figure 3 shows that the average annual citation numbers exhibit a fluctuating downward trend during the 2011–2025 period, and this is due to bibliometric factors such as the expansion of the literature and the fact that publications are still in the citation collection phase. Older studies have had more time to accumulate citations, whereas more recent publications, especially those published after 2021, are still in the citation accumulation phase. For this reason, the decline or fluctuation in average citations does not necessarily indicate a decrease in scientific impact. Rather, it reflects the rapid expansion of a young and dynamic literature in which many recent studies have not yet reached their full citation potential.

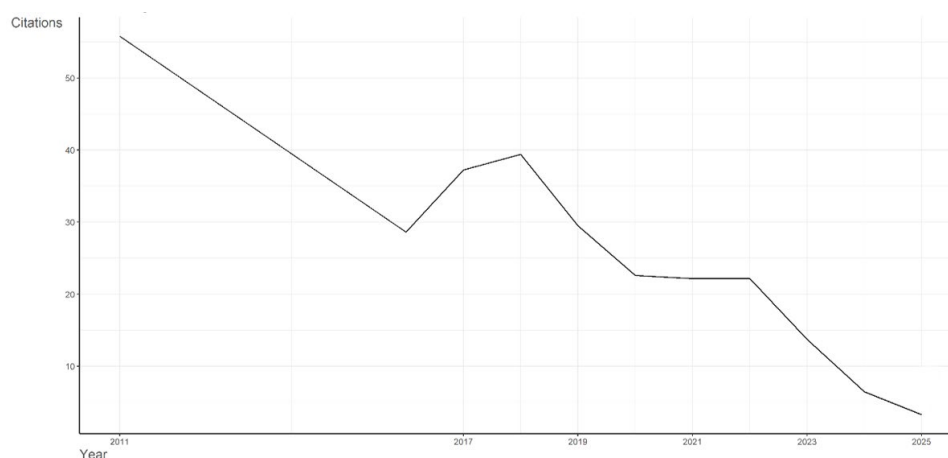


Figure 3. Annual average citations

The three-field graph (Figure 4) comprehensively demonstrates the structural relationship between the intellectual foundations, prominent authors, and dominant themes of the literature. It is evident that the cited key works are based on empirical and econometric research focusing on the energy-economy-environment relationship; particularly foundational works, such as those by Ozturk & Acaravci (2013), Sadorsky (2010), Tamazian & Rao (2010), and Zhang (2011), have shaped the literature along the axes of financial development, energy consumption, economic growth, and environmental quality. An examination of the author network reveals contributions from Paramati et al. (2017), Shahbaz et al. (2022), and Sinha & Shahbaz (2018). It is understood that researchers, such as Ahmad et al. (2020), Irfan et al. (2022a), and Sharif et al. (2020), played a central role in the development of the field and that there is a strong international collaboration structure. The keywords show that the core of the literature was shaped around “financial development,” “renewable energy,” “carbon emissions,” and “economic growth,” and in recent years it has evolved into a broader and sustainability-oriented framework with concepts such as “green finance,” “environmental sustainability,” “technological innovation,” and “institutional quality”.

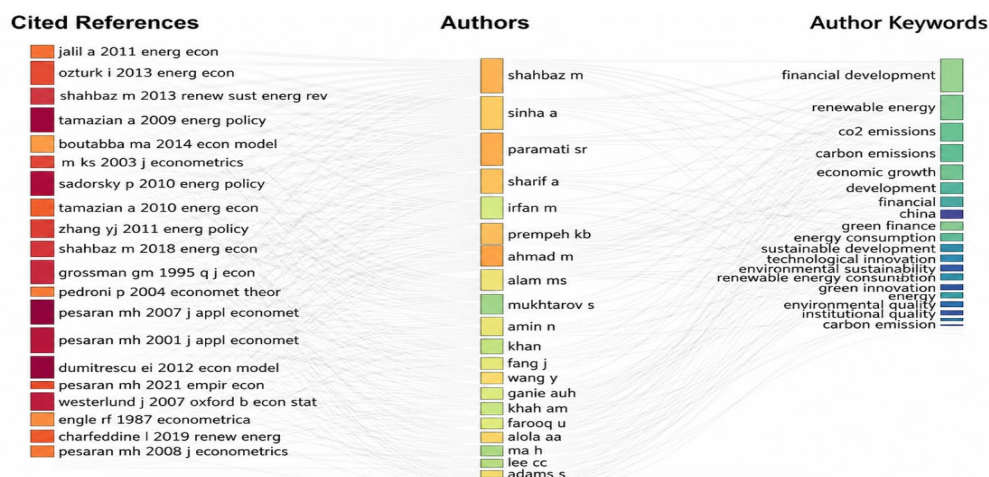


Figure 4. Three-field graph

Figure 5 shows that the studies are predominantly published in journals focusing on energy economics, sustainability policies, and environmental economics. The journal with the most publications is Energy Economics ($n = 29$), revealing that the literature addresses the energy-finance-environment relationship through econometric and policy-based analyses. This is followed by the Journal of the Knowledge Economy ($n = 17$) and the International Journal of Energy Sector Management ($n = 14$); thus, it is understood that the field is addressed not only from the perspective of energy economics but also from the perspectives of innovation, knowledge economy, and energy management. Energy Policy and Technological Forecasting and Social Change ($n = 13$) strengthen the policy and technological transformation dimension, while Economic Research–Ekonomiska Istraživanja ($n = 12$), Economic Analysis and Policy ($n = 10$), and Economic Change and Restructuring ($n = 8$) represent the macroeconomic and structural transformation axis. Applied Economics and Cogent Economics and Finance ($n = 7$) show that applied econometric studies hold a significant place in the literature. Overall, the graph reveals that the field has an interdisciplinary but strongly economics- and energy-based publication structure.

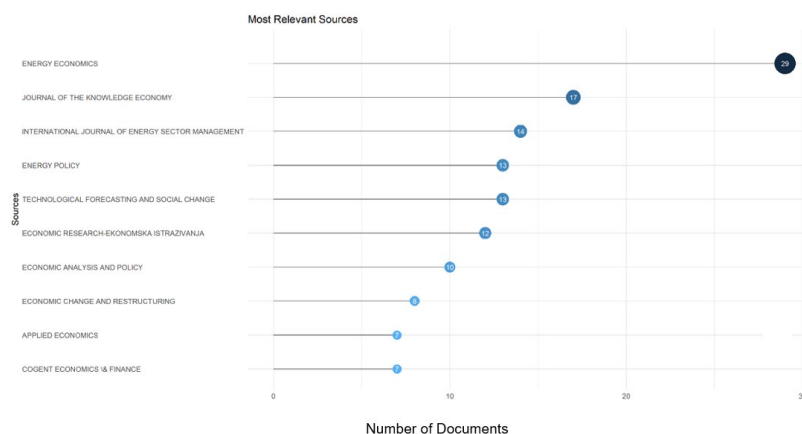


Figure 5. Most relevant resources

Figure 6 shows that the publication performance of the most relevant journals in financial development, renewable energy, and environmental sustainability has increased rapidly, especially since 2017. The most significant increase was observed in the journal *Energy Economics*, which gained momentum after 2021 and reached its highest cumulative output by 2025. While *Energy Policy* showed a steady increase, the *International Journal of Energy Sector Management* and the *Journal of the Knowledge Economy* stood out with their accelerated contributions, especially in the post-2020 period. *Technological Forecasting and Social Change*, on the other hand, gained momentum between 2019 and 2023 and made significant contributions to studies on technological transformation. Overall, the graph reveals that the literature expanded after 2017, accelerated significantly after 2020, and that the field, while centered on energy economics, is increasingly multidisciplinary, encompassing technology, policy, and knowledge economics.

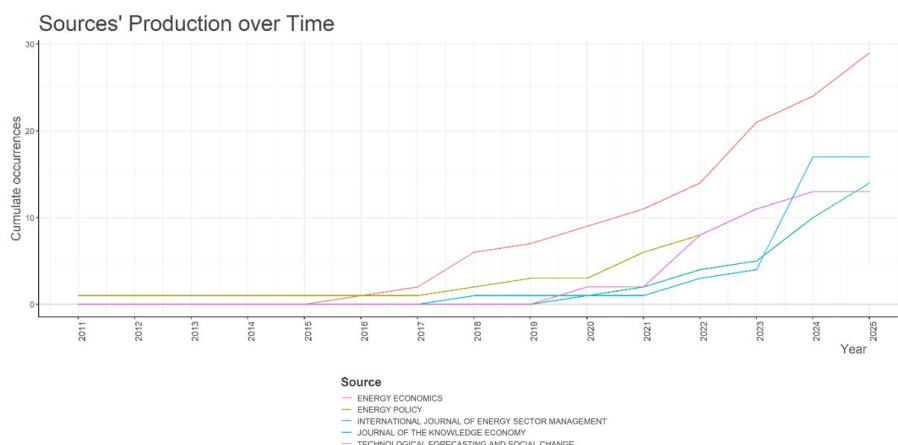


Figure 6. Annual cumulative production amount of resources

Figure 7 shows the most prolific researchers in the literature on financial development, renewable energy, and environmental sustainability. Some authors have established a central position in the literature, particularly by examining the relationships between financial development, renewable energy, carbon emissions, and sustainable development through econometric and policy-oriented analyses. Some have made significant contributions in the areas of renewable energy, climate change, technological innovation, and sustainability, while others focus on the relationships between environmental footprint, tourism, and growth. Some contribute to the fields of energy policies and economic transformation. Overall, the graph shows that the literature is concentrated around a specific core group of researchers and that the field has matured thematically and methodologically, becoming an interdisciplinary structure.

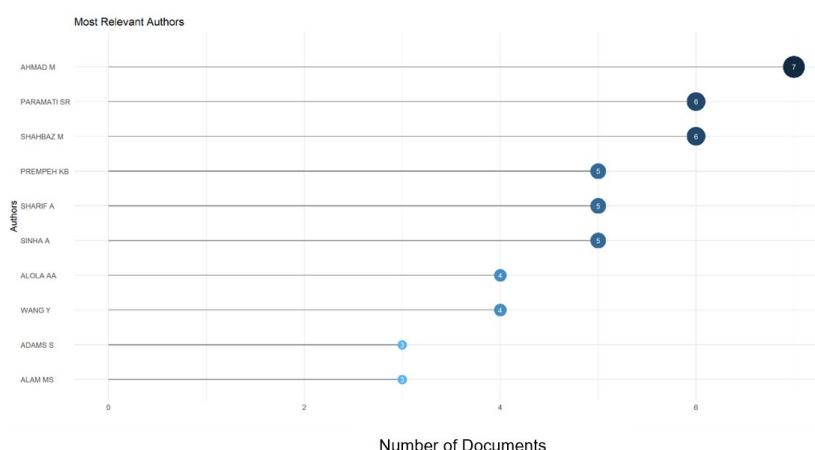


Figure 7. Most relevant authors

Figure 8 reveals the publication performance and citation impact of prominent researchers over the years, showing a strong acceleration in the literature, particularly after 2017 and significantly after 2020. Ahmad et al. (2020) stand out with his increased publication visibility and citation impact after 2020, while Paramati et al. (2017) and Shahbaz et al. (2020) demonstrate a stable and influential position in the literature by spreading their output

over a longer period. The increased contributions of authors, especially in the post-2020 period, show that the visibility of studies on sustainability, corporate quality, and the energy-environment relationship has increased. Similarly, researchers also contribute thematic diversity to the literature. Overall, the graph reveals that output is concentrated around a specific core group of authors, and that studies on the finance-energy-environment axis have strengthened both in quantity and impact, particularly after 2018 and more significantly after 2020.

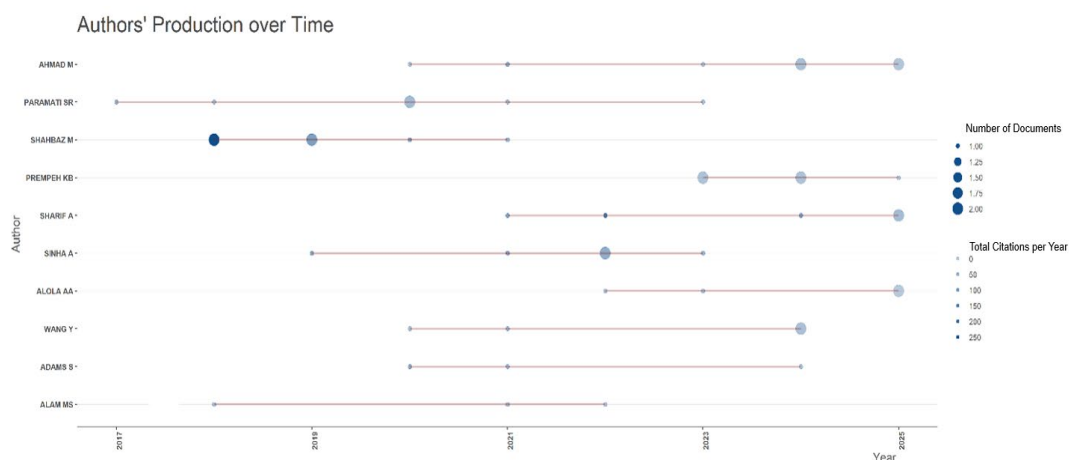


Figure 8. Author's annual production quantity

Figure 9 shows the distribution of publications by country of responsible authors and the ratios of single-country publications and multinational collaborative publications, revealing that while the literature is geographically broad, it is concentrated in certain countries. China is by far the country with the highest output, leading in both single-country publications and multinational collaborative publications, and is also very active in international collaborations. China is followed by India and Pakistan; India's high multinational collaborative publications ratio, in particular, indicates strong international integration. Ghana, Vietnam, and Turkey exhibit moderate but increasing output, with relatively high single-country publication ratios and strengthening international collaborations in these countries. Australia, France, and Saudi Arabia stand out more for multinational studies, while the United Kingdom, Malaysia, South Africa, and the United Arab Emirates have a more balanced output structure. The relatively low output of the United States of America can be explained by the fact that the field is addressed in that country under different thematic headings in a scattered manner.

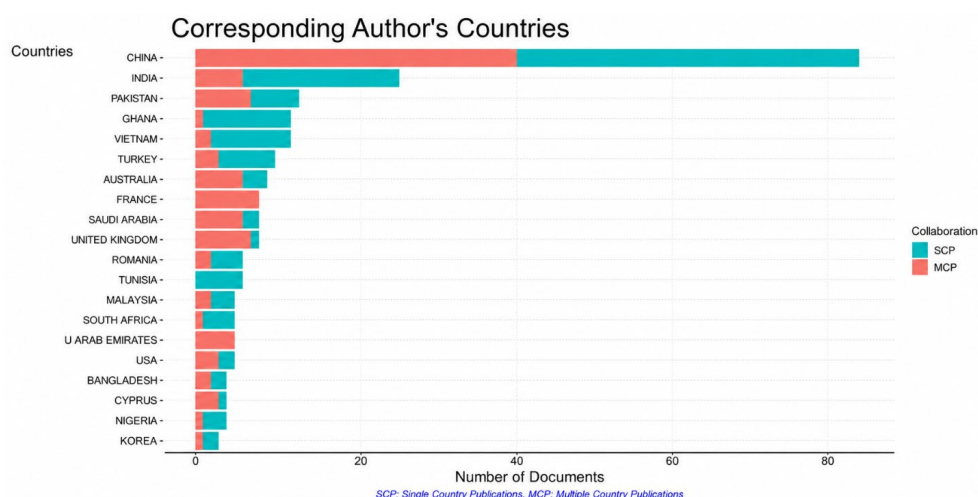


Figure 9. Chart of corresponding author countries

Figure 10 illustrates the geographical distribution of scientific output in the research area and shows that the literature is distinctly Asia-centric. China stands out with by far the highest number of publications, while South Asian countries such as Pakistan and India also hold a significant place in the literature with studies addressing the relationship between economic growth, carbon emissions, and renewable energy. Vietnam, Malaysia, and some other developing countries are notable for their increasing empirical contributions, while in Europe, the United

Kingdom and France provide methodological depth primarily through studies in the areas of energy policies, carbon markets, and sustainable finance. Contributions from Saudi Arabia in the Middle East, Australia, and some African countries (e.g., Ghana) demonstrate the diversification of the literature on a global scale. Overall, the findings confirm that environmental sustainability and renewable energy issues are gaining increasing importance on the academic and policy agendas, particularly in developing countries.

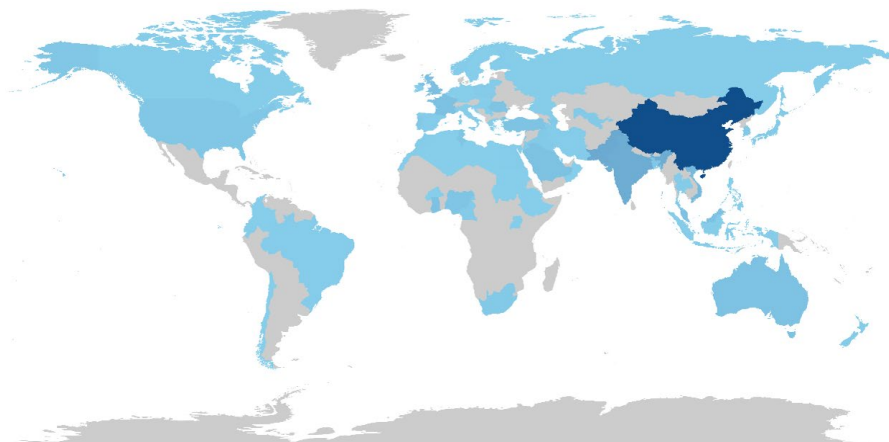


Figure 10. Amount of scientific output by countries

Figure 11 shows that the scientific output of selected countries increased significantly after 2017, but growth was much faster, especially in China. China's output accelerated after 2018, rising sharply from 2021 onwards to over 300 in 2025, significantly widening the gap with other countries. In India, Pakistan, and Vietnam, a more gradual but steady increase is observed; the rise accelerates particularly after 2021. The United Kingdom exhibits more limited but consistent output. Overall, the graph reveals that growth in the literature is largely Asia-centric, with China achieving a dominant academic position, particularly in the areas of renewable energy, environmental sustainability, and financial development.

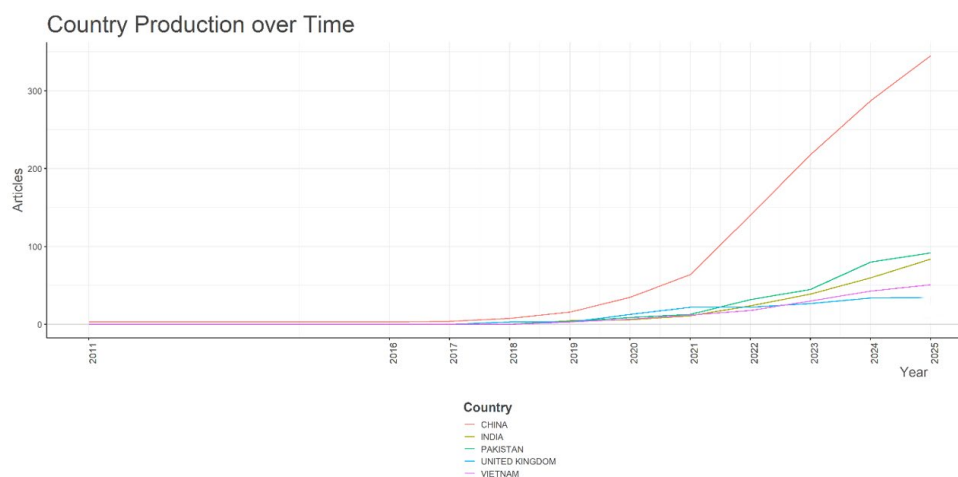


Figure 11. Annual production amount of countries

Figure 12 shows that the most cited studies globally in the literature are articles examining the energy-finance-environment relationship within an empirical and econometric framework. The most cited study is Balsalobre-Lorente et al. (2018) (Energy Policy, $n = 1148$), which comprehensively addresses the relationship between economic growth, renewable energy, and carbon emissions. It reveals that the interactions between financial development, energy consumption, and environmental quality have become key references in the literature. Other studies by Irfan et al. (2022b), Ji & Zhang (2019), and Shahbaz et al. (2020) also stand out with their high citation impact, while authors such as Avom et al. (2020), Aye & Edoja (2017), and Khan et al. (2022) contribute to more recent sub-themes. Overall, the graph shows that high-impact studies are predominantly published in journals such as Energy Policy, Energy Economics, and Technological Forecasting and Social Change, and that the field is shaped around empirical energy-finance-environment analyses.

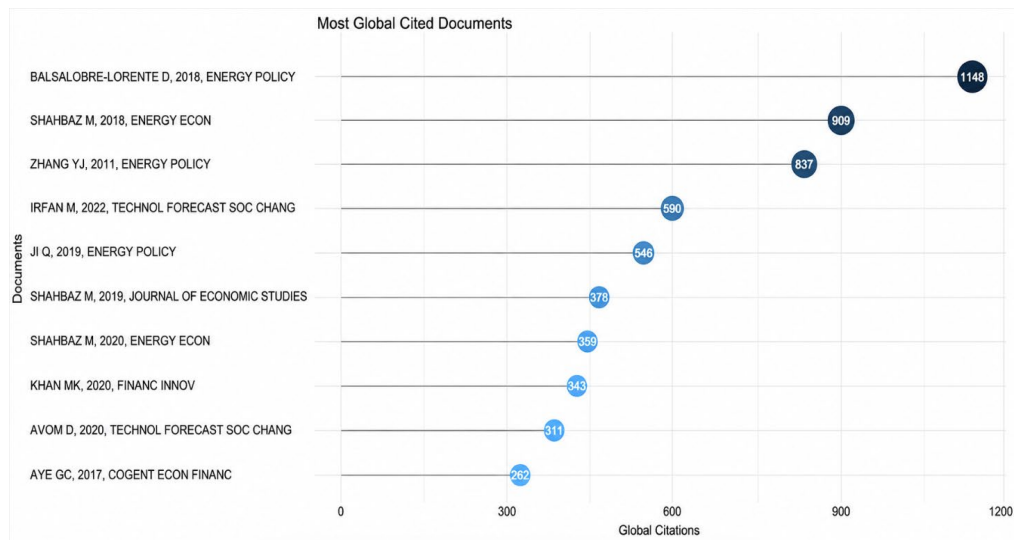


Figure 12. Most cited works

Figure 13 shows that the most frequently used Keywords Plus terms in the literature are “CO₂ emissions” ($n = 169$) and “economic growth” ($n = 144$), indicating that studies primarily focus on the relationship between economic growth, energy consumption, and environmental degradation. The high frequency of the terms “renewable energy” ($n = 104$) and “financial development” ($n = 87$) indicates increasing interest in the impact of the energy transition and financial systems on sustainability. Concepts such as “impact” ($n = 86$), “consumption,” “energy consumption,” “growth,” “foreign direct investment,” and “cointegration” reflect both thematic diversity and the prevalence of econometric analyses examining long-term relationships. Overall, the graph shows that the literature is concentrated along the energy-finance-growth-environment axis, with discussions of carbon emissions and sustainable development particularly central.

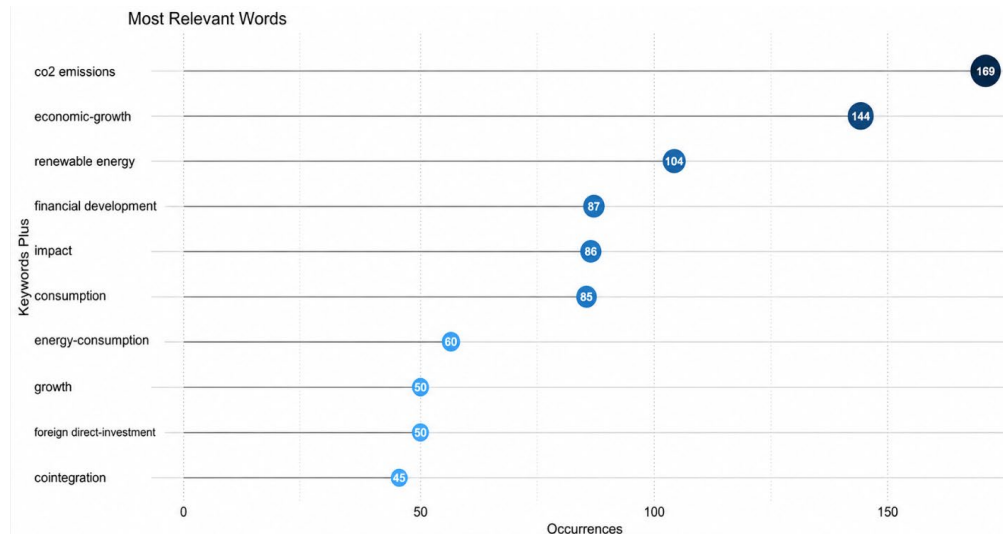


Figure 13. Word frequency

Figure 14 clearly shows that the literature’s thematic focus is on the concept of “CO₂ emissions” and that research is largely shaped by environmental degradation and carbon emissions. The prominence of the terms “economic growth,” “renewable energy,” and “financial development” reveals that the energy-finance-growth relationships are considered together within the framework of sustainability. Concepts such as “consumption,” “energy consumption,” “trade openness,” “foreign direct investment,” “cointegration,” and “panel data,” which are moderately prominent, reflect both the thematic diversity and the widespread use of econometric methods. Furthermore, terms such as “Environmental Kuznets Curve,” “urbanization,” “electricity consumption,” and “climate change” demonstrate that the literature integrates with broader macro themes, including climate change, global integration, and structural transformation.



Figure 14. Word cloud

The keyword-based findings indicate that the field has moved from a relatively narrow focus on the relationship between economic growth, energy consumption, and CO₂ emissions toward a broader sustainability-oriented framework. The dominance of terms such as “CO₂ emissions,” “economic growth,” “renewable energy,” and “financial development” shows that the core of the literature is still shaped by the energy–economy–environment nexus. However, the growing visibility of concepts such as “green finance,” “technological innovation,” “foreign direct investment,” “trade openness,” and “environmental sustainability” suggests that the research field has gradually incorporated macro-financial and institutional dimensions.

Figure 15 shows that the most frequently used keywords have gained significant momentum since 2017, indicating a thematic shift in the literature. The fastest increase is seen in the term “CO₂ emissions”, while “economic growth” and “renewable energy” also exhibit a strong and sustained rise, revealing an intensification of studies focusing on the energy-economy-environment relationship. “Financial development,” “impact,” and “energy consumption” show a steady increase, reflecting the growing importance of sustainable finance and energy consumption themes. “Foreign direct investment,” “consumption,” and “cointegration” exhibit a more gradual but continuous usage trend, confirming both the thematic diversity and the prevalence of econometric methods in the literature. Overall, the graph shows that the literature has rapidly expanded over the last five years and evolved into a multidisciplinary structure centered on energy transition and environmental sustainability.

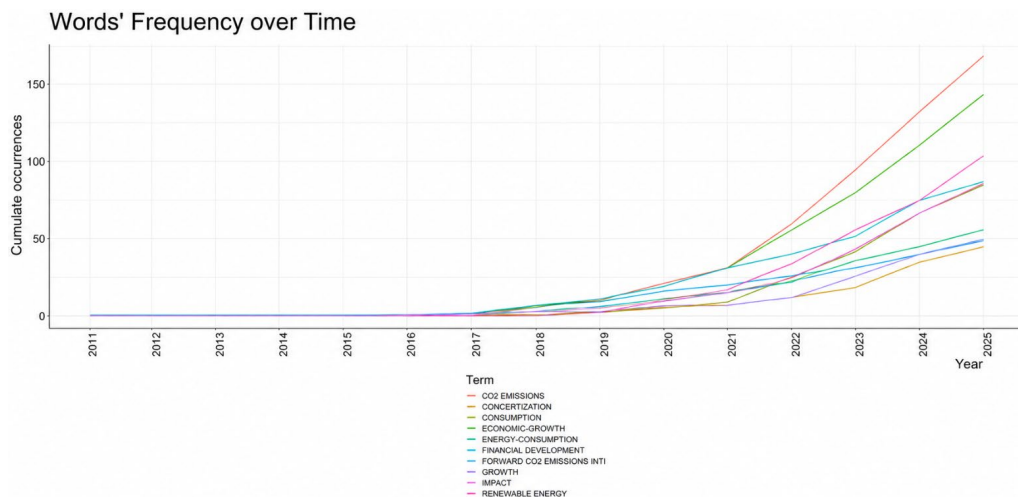


Figure 15. Word frequency over the years

Figure 16 shows that there has been a significant thematic expansion in the literature, especially after 2021, and that research has mainly focused on “CO₂ emissions,” “economic growth,” and “renewable energy.” The high frequency of these concepts reveals that environmental sustainability and energy transition have taken a central place in the literature. The prominence of global integration indicators such as “foreign direct investment,” “trade

openness,” and “international trade” in the post-2021 period shows that the international trade dimension has been integrated into the energy-environment relationship. The resurgence of the “environmental Kuznets curve” and related concepts after 2023 indicates a revival of theoretical discussions on the growth-environment relationship. While terms related to energy consumption attracted attention during the 2020–2022 period, it is observed that the concept of “clean energy,” used earlier, has gradually been replaced by the more comprehensive term “renewable energy.”

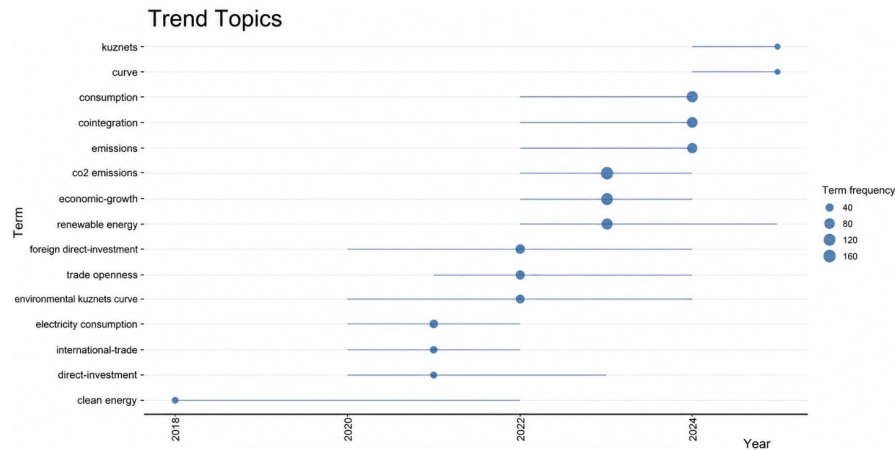


Figure 16. Trending topics

This thematic shift is particularly evident in the post-2020 period. The increasing use of terms related to renewable energy, green finance, and sustainable development shows that the literature has begun to focus more strongly on transition mechanisms rather than merely identifying the environmental consequences of economic growth. In this respect, the field appears to have evolved from an early stage dominated by growth–emission debates toward a more mature stage in which financial instruments, innovation, international integration, and energy transition are examined together.

Figure 17 shows that the most frequently used concepts in the literature are grouped around two main thematic clusters. The red cluster represents studies based on the traditional energy-economy-environment relationship, and especially the Environmental Kuznets Curve framework, with concepts such as economic growth, CO₂ emissions, financial development, foreign direct investment, trade openness, and energy consumption. The blue cluster reflects newer research trends focused on energy transformation, sustainability, and technological innovation with concepts such as renewable energy, consumption, impact, environmental degradation, and innovation. While the large nodes in the center (CO₂ emissions, economic growth, renewable energy, financial development) form the backbone of the field, the strong connections between the clusters demonstrate a robust integration between the traditional growth-environment literature and studies based on renewable energy and sustainability. The connection between these two clusters indicates that the field is not fragmented; rather, it is developing through the integration of traditional macroeconomic and environmental approaches with newer themes such as renewable energy transition, technological innovation, and green finance.

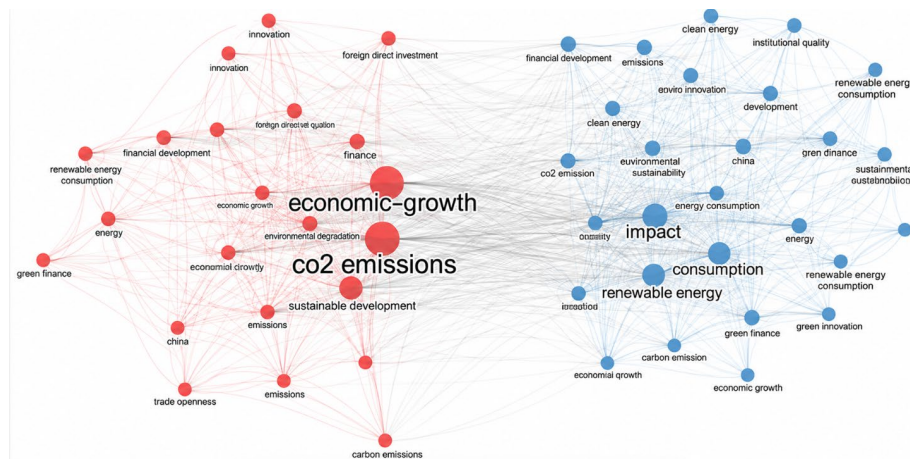


Figure 17. Keyword co-occurrence network

Figure 18 shows that international partnerships in the research area are particularly concentrated in Asia and Europe, with China being the most central actor in the network. China's strong connections with numerous countries demonstrate its position as a global hub in energy economics and environmental sustainability studies. Asian countries such as India, Pakistan, Vietnam, and Malaysia also play an active role in the cooperation network, while in Europe, the United Kingdom, France, Romania, and Turkey stand out as important connection points. Australia's strong relationships with both Asia and Europe support global academic integration. The contribution of Africa and the Americas is more limited but is on an upward trend. Overall, the map shows that the energy-finance-environment literature has a highly internationalized, multi-centric, but predominantly Asia-centric cooperation structure.

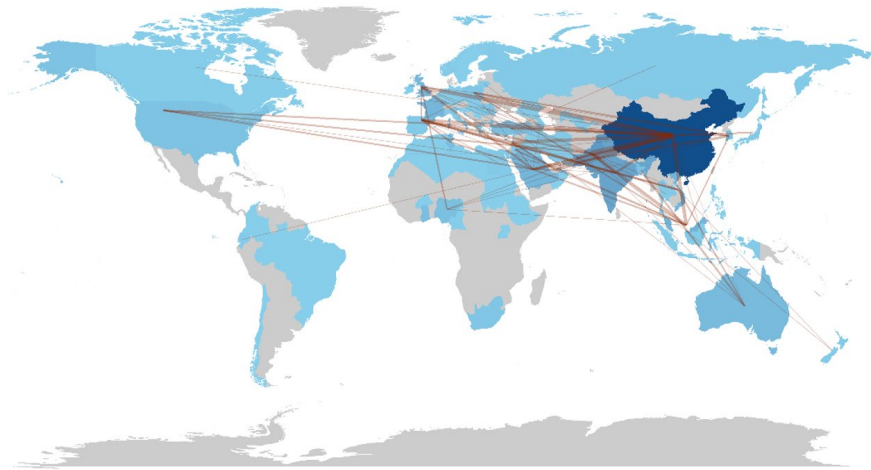


Figure 18. Country partnerships map

The country-based findings show that the geographical development of the literature is strongly concentrated in emerging and developing economies, particularly China, India, and Pakistan. This pattern can be explained by the fact that these countries face simultaneous challenges of economic growth, rising energy demand, environmental degradation, and the need for sustainable financing. Therefore, their leading position in scientific production reflects not only academic capacity but also the policy relevance of the research topic in countries where growth and environmental pressures are closely linked. The collaboration patterns also indicate that the literature has become increasingly internationalized. The high level of multinational collaboration suggests that carbon emissions, renewable energy transition, and green finance are no longer treated as country-specific issues but as global research problems requiring cross-country data, comparative methods, and international academic networks. The central position of China and the increasing contribution of countries such as India, Pakistan, Vietnam, Turkey, and Ghana show that the field is becoming more diversified, although it remains largely Asia-centered.

4. Discussion

This bibliometric study examines the development of academic literature on financial development, ecological footprint, renewable energy, and the impact of green finance on CO₂ emissions between 2011 and 2025. By analyzing 295 studies from the Web of Science database, the structure of the field, its thematic evolution, scientific collaboration networks, and the most influential publications and researchers are systematically revealed. The findings confirm the increasing importance of these issues in both finance literature and sustainable development policies. The study's findings show that discussions in the literature largely focus on the multifaceted relationships between renewable energy use, financial development, energy consumption, economic growth, and CO₂ emissions. The prominence of key concepts such as economic growth, CO₂ emissions, financial development, and renewable energy confirms that the literature centers on the search for a balance between economic development, financial advancement, and environmental sustainability.

Word cloud analysis, trend analysis, and co-occurrence network results reveal that the research theme has rapidly diversified in the post-2020 period, with concepts such as renewable energy, international trade openness, foreign direct investment, and financial globalization being increasingly studied. This indicates that the literature has become sensitive not only to environmental factors but also to the dynamics of economic and financial integration. Country and author analyses of the study show that China, India, and Pakistan exhibit clear leadership in scientific output; the intensity of research in these countries is consistent with both population size and increasing environmental pressures. The most prolific authors have made significant contributions, shaping theory

and practice in the fields of energy economics and sustainable development. Citation analyses indicate that the impacts of financial development on environmental quality are strongly debated in the international literature, particularly in studies such as Balsalobre-Lorente et al. (2018) and Shahbaz et al. (2020).

Thematic maps and trend analyses show that the field has moved from the exploration phase to maturity, and that new research axes such as green finance, carbon neutrality, energy transition, and sustainable financing are gaining momentum. It is thought that financing renewable energy investments, the capacity of financial markets to price environmental risks, and the development of green bond markets could be key areas of future research.

5. Conclusions

In conclusion, the bibliometric findings obtained within the scope of this study reveal that the relationship between financial development, renewable energy, and environmental indicators has become a significant research area in the literature. Existing studies indicate that the functioning of financial systems and energy policies can affect environmental performance through different channels. In this context, it is observed that considering economic growth and environmental sustainability goals together is gaining increasing importance in the literature. This study presents the general trends and conceptual structure of the field from a holistic perspective, shedding light on potential thematic directions for future research. It is considered that the findings can contribute to identifying gaps in the literature for researchers and enriching discussions on sustainable development goals for policymakers. The findings indicate that achieving environmental sustainability goals requires consideration not only of energy policies but also of the structure and functioning of financial systems. General trends in the literature show that the widespread adoption of sustainable financial instruments, the encouragement of green investments, and the regulation of financial markets to more effectively reflect environmental risks are important for policymakers. In this context, aligning the level of financial development with environmental goals can be considered a complementary element in supporting long-term sustainable development strategies.

The study also has some limitations. First, the analysis is limited to publications included in the Web of Science database. Secondly, while the bibliometric analysis method is powerful in revealing the structural and thematic characteristics of the literature, it does not allow for direct inferences about causal relationships. Finally, keyword-based analyses can be affected by author preferences and indexing differences, so the relative weight of some themes may appear higher or lower than it actually is. Despite these limitations, the study provides a comprehensive overview of the relationship among financial development, renewable energy, and environmental indicators in the literature and has the potential to serve as a guiding reference for future empirical and theoretical studies. This bibliometric study contributes to the literature by presenting a holistic perspective on the multidimensional relationship between financial development, renewable energy, and CO₂ emissions using an up-to-date dataset covering the period 2011–2025. Similar to existing literature, it centers on the search for a balance between economic growth and environmental sustainability, validating the theoretical frameworks of pioneering authors. However, the key difference that sets this study apart is its empirical proof of the weight of new research themes that emerged after 2020, such as “green finance,” “carbon neutrality,” and “green bonds”.

In terms of future research directions, the findings suggest that the literature should move beyond examining the general relationship between financial development and CO₂ emissions and focus more specifically on the channels through which financial systems affect environmental outcomes. Future studies may examine the roles of green bonds, Environmental, Social, and Governance-based investment, sustainable banking, climate-related financial regulation, and renewable energy financing as mechanisms linking financial development to carbon reduction. Moreover, the growing importance of technological innovation, trade openness, foreign direct investment, and financial globalization indicates that future research should consider the international and institutional dimensions of the energy–finance–environment nexus. From a policy perspective, the findings imply that environmental sustainability requires not only energy policies but also financial policies that support green investment, improve climate-related disclosure, and encourage the allocation of capital toward low-carbon technologies.

Author Contributions

Conceptualization, H.A. and H.E.K.; methodology, H.A. and H.E.K.; software, H.A.; validation, H.A. and H.E.K.; formal analysis, H.A. and H.E.K.; investigation, H.A. and H.E.K.; resources, H.A. and H.E.K.; data curation, H.A.; writing—original draft preparation, H.A. and H.E.K.; writing—review and editing, H.A. and H.E.K.; visualization, H.A.; supervision, H.E.K.; project administration, H.A. and H.E.K. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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